ENR 3.2 UPPER ATS ROUTES

Route designator *Name of significant points* *Coordinates*	*Route* *MAG* *Track* *MAG* *DIST*	*Upper limit* *Lower limit* *Classification*	*Lateral limits* *NM*	*Direction of cruising levels*		*Remarks* *Controlling unit* *Frequency* *SATVOICE* *Logon address*
				Odd	*Even*	
1	*2*	*3*	*4*	*5*		*6*
UA402						
▲CAPE TOWN INTERNATIONAL AIRPORT (CTV) VOR/DME 335810.82S 0183618.19E						
	126° 306° 44 NM	FL 280 FL 245 Class A	10	↓	↑	
▲CALEDON (CDV) VOR 340646.68S 0192840.39E						
	113° 293° 145 NM	FL 280 FL 245 Class A	10	↓	↑	
▲GEORGE AIRPORT (GRV) VOR/DME 340026.46S 0222233.46E						
	117° 296° 161 NM	FL 280 FL 245 Class A	10	↓	↑	
▲CHIEF DAWID STUURMAN AIRPORT (PEV) VOR/DME 335933.01S 0253620.75E						
	091° 270° 125 NM	FL 280 FL 245 Class A	10	↓	↑	
▲KING PHALO AIRPORT (ELV) VOR/DME 330213.16S 0274923.36E						
	073° 251° 121 NM	FL 280 FL 245 Class A	10	↓	↑	
▲PORT ST JOHNS (PJV) VOR 313631.96S 0293111.49E						FAJA/FACA FIR BDRY
	062° 240° 96 NM	FL 280 FL 245 Class A	10	↓	↑	

Route designator *Name of significant points* *Coordinates*	*Route* *MAG* *Track* *MAG* *DIST*	*Upper limit* *Lower limit* *Classification*	*Lateral limits* *NM*	*Direction of cruising levels*		*Remarks* *Controlling unit* *Frequency* *SATVOICE* *Logon address*
				Odd	*Even*	
1	*2*	*3*	*4*	*5*		*6*
▲OKTAN 301758.32S 0303453.34E						
	060° 239° 50 NM	FL 280 FL 245 Class A	10	↓	↑	
▲KING SHAKA (TGV) VOR/DME 293640.20S 0310729.03E						
	117° 297° 50 NM	FL 460 FL 280 Class A	10	↓	↑	
▲EVIDO 293844.82S 0320448.90E						
	094° 273° 59 NM	FL 460 FL 280 Class A	10	↓	↑	
▲ESRET 291639.66S 0330805.46E						
	095° 275° 72 NM	FL 460 FL 280 Class A	10	↓	↑	
▲APLUK 285213.00S 0342532.00E						FAJA/FAJO FIR BDRY
	095° 274° 85 NM	FL 460 FL 280 Class A	10	↓	↑	
▲ETMOS 282224.00S 0355618.00E						FAJO/FQBE FIR BDRY

UA405						
▲GREEFSWALD (GWV) VOR 221444.60S 0292437.21E						FAJA/FVHF FIR BDRY
	215° 038° 124 NM	FL 280 FL 245 Class A	10	↑	↓	
▲GABNA 241000.00S 0283400.00E						
	219° 041° 99 NM	FL 280 FL 245 Class A	10	↑	↓	
▲UTRUK 254032.98S 0274957.29E						

Route designator *Name of significant points* *Coordinates*	*Route* *MAG* *Track* *MAG* *DIST*	*Upper limit* *Lower limit* *Classification*	*Lateral limits* *NM*	*Direction of cruising levels*		*Remarks* *Controlling unit* *Frequency* *SATVOICE* *Logon address*
				Odd	*Even*	
1	*2*	*3*	*4*	*5*		*6*
	207° 028° 51 NM	FL 280 FL 245 Class A	10	↑	↓	
▲GRASMERE (GAV) VOR 263053.63S 0274035.17E						
	248° 069° 135 NM	FL 280 FL 245 Class A	10	↑	↓	
▲EKBEX 280039.10S 0254723.96E						
	249° 069° 28 NM	FL 280 FL 245 Class A	10	↑	↓	
▲APDUR 281919.65S 0252312.80E						
	249° 070° 44 NM	FL 280 FL 245 Class A	10	↑	↓	
▲KIMBERLEY AIRPORT (KYV) VOR/DME 284759.82S 0244537.32E						
	229° 052° 162 NM	FL 280 FL 245 Class A	10	↑	↓	
▲EGRUP 311005.95S 0231539.78E						FAJA/FACA FIR BDRY
	233° 053° 16 NM	FL 280 FL 245 Class A	10	↑	↓	
▲VICTORIA WEST (VWV) VOR 312349.00S 0230632.00E						
	269° 088° 142 NM	FL 280 FL 245 Class A	10	↑	↓	
▲SUTHERLAND (SLV) VOR 322700.00S 0203721.00E						
	255° 076° 77 NM	FL 280 FL 245 Class A	10	↑	↓	
▲CERES (CSV) VOR/DME 331508.27S 0192617.22E						

Route designator *Name of significant points* *Coordinates*	*Route* *MAG* *Track* *MAG* *DIST*	*Upper limit* *Lower limit* *Classification*	*Lateral limits* *NM*	*Direction of cruising levels*		*Remarks* *Controlling unit* *Frequency* *SATVOICE* *Logon address*
				Odd	*Even*	
1	*2*	*3*	*4*	*5*		*6*
	248° 069° 60 NM	FL 280 FL 245 Class A	10	↑	↓	
▲CAPE TOWN INTERNATIONAL AIRPORT (CTV) VOR/DME 335810.82S 0183618.19E						

UB529						
▲KING SHAKA (TGV) VOR/DME 293640.20S 0310729.03E						
	070° 250° 50 NM	FL 460 FL 245 Class A	10	↓	↑	
▲ITMIL 290151.05S 0314838.40E						
	064° 244° 23 NM	FL 460 FL 245 Class A	10	↓	↑	
▲RICHARDS BAY (RBV) VOR/DME 284426.49S 0320536.91E						
	032° 210° 114 NM	FL 460 FL 245 Class A	10	↓	↑	
▲EPRET 265131.86S 0322546.72E						FAJA/FQBE FIR BDRY

UB534						
▲WITBANK (WIV) VOR 254946.29S 0291141.13E						
	128° 309° 91 NM	FL 280 FL 245 Class A	10	↓	↑	
▲DUKRA 262024.43S 0304711.57						FAJA/FDMS TMA BDRY

Route designator *Name of significant points* *Coordinates*	*Route* *MAG* *Track* *MAG* *DIST*	*Upper limit* *Lower limit* *Classification*	*Lateral limits* *NM*	*Direction of cruising levels*		*Remarks* *Controlling unit* *Frequency* *SATVOICE* *Logon address*
				Odd	*Even*	
1	*2*	*3*	*4*	*5*		*6*
	149° 330° 27 NM	FL 280 FL 245 Class A	10	↓	↑	
▲MUVAL 263728.00S 0310957.00E						

UB540						
UTRUK 254032.98S 0274957.29E						
	013° 190° 148 NM	FL 460 FL 200 Class A	10	↓	↑	
ETMIT 231248.49S 0273742.69E						FAJA/FBGR FIR BDRY

UG465						
▲OKTEL 280753.81S 0150000.00E						FAJA/FAJO FIR BDRY
	127° 306° 85 NM	FL 280 FL 245 Class A	10	↓	↑	
▲ALEXANDER BAY (ABV) VOR/DME 283414.35S 0163201.80E						
	108° 286° 236 NM	FL 280 FL 245 Class A	10	↓	↑	
▲IBVIT 282453.62S 0210000.44E						
	106° 286° 14 NM	FL 280 FL 245 Class A	10	↓	↑	
▲UPINGTON AIRPORT (UPV) VOR/DME 282406.31S 0211537.48E						

Route designator *Name of significant points* *Coordinates*	*Route* *MAG* *Track* *MAG* *DIST*	*Upper limit* *Lower limit* *Classification*	*Lateral limits* *NM*	*Direction of cruising levels*		*Remarks* *Controlling unit* *Frequency* *SATVOICE* *Logon address*
				Odd	*Even*	
1	*2*	*3*	*4*	*5*		*6*
	083° 263° 103 NM	FL 280 FL 245 Class A	10	↓	↑	
▲SISHEN (SSV) VOR/DME 273907.83S 0230010.99E						
	087° 266° 283 NM	FL 280 FL 245 Class A	10	↓	↑	
▲LANSERIA AIRPORT (LIV) VOR/DME 255656.14S 0275448.86E						
	102° 281° 70 NM	FL 280 FL 245 Class A	10	↓	↑	
▲WITBANK (WIV) VOR 254946.29S 0291141.13E						
	059° 240° 95 NM	FL 280 FL 245 Class A	10	↓	↑	
▲ANVIT 244000.00S 0302300.00E						
	058° 238° 62 NM	FL 280 FL 245 Class A	10	↓	↑	
▲PHALABORWA (PHV) VOR 235402.73S 0310907.09E						
	055° 235° 35 NM	FL 280 FL 245 Class A	10	↓	↑	.
▲EPSEK 232715.00S 0313345.00E						FAJA/FQBE FIR BDRY
Subject to FUA approval through military airspace.						

UG467						
▲KRUGER MPUMALANGA (PKV) VOR 252251.69S 0310640.05E						
	069° 249° 20 NM	FL 280 FL 245 Class A	10	↓	↑	

Route designator *Name of significant points* *Coordinates*	*Route* *MAG* *Track* *MAG* *DIST*	*Upper limit* *Lower limit* *Classification*	*Lateral limits* *NM*	*Direction of cruising levels*		*Remarks* *Controlling unit* *Frequency* *SATVOICE* *Logon address*
				Odd	*Even*	
1	*2*	*3*	*4*	*5*		*6*
▲DUSKO 251028.74S 0312401.70E						
	069° 248° 42 NM	FL 280 FL 245 Class A	10	↓	↑	
▲TEVAP 244430.00S 0320006.00E						FAJA/FQBE FIR BDRY

UG653						
▲ETOSA 244250.40S 0260312.98E						FAJA/FGBR FIR BDRY
	136° 317° 37 NM	FL 280 FL 195 Class A	10	↓	↑	
▲NESEK 250154.00S 0263700.00E						
	137° 316° 77 NM	FL 280 FL 195 Class A	10	↓	↑	
▲UTRUK 254032.98S 0274957.29E						

UG655						
▲MASERU (MZV) VOR 293001.00S 0273226.00E						
	036° 212° 172 NM	FL 280 FL 245 Class A	10	↓	↑	
▲HEIDELBERG (HGV) VOR 264148.50S 0281659.39E						
	357° 176° 66 NM	FL 280 FL 245 Class A	10	↑	↓	
▲UTRUK 254032.98S 0274957.29E						

Route designator *Name of significant points* *Coordinates*	*Route* *MAG* *Track* *MAG* *DIST*	*Upper limit* *Lower limit* *Classification*	*Lateral limits* *NM*	*Direction of cruising levels*		*Remarks* *Controlling unit* *Frequency* *SATVOICE* *Logon address*
				Odd	*Even*	
1	*2*	*3*	*4*	*5*		*6*
	026° 204° 93 NM	FL 280 FL 245 Class A	10	↓	↑	
▲ODKER 240830.00S 0280500.00E						
	023° 200° 92 NM	FL 280 FL 245 Class A	10	↓	↑	
▲TAVLA 223724.00S 0281736.00E						FAJA/FBGR FIR BDRY

UG853						
▲KING SHAKA (TGV) VOR/DME 293640.20S 0310729.03E						
	291° 111° 38 NM	FL 280 FL 245 Class A	10	↑	↓	
▲PIETERMARITZBURG (PMV) VOR/DME 293852.13S 0302401.28E						
	353° 172° 73 NM	FL 280 FL 245 Class A	10	↑	↓	
▲LADYSMITH (LYV) VOR/DME 283616.21S 0294140.32E						
	349° 167° 60 NM	FL 280 FL 245 Class A	10	↑	↓	
▲WARDEN (WRV) VOR 274629.26S 0290410.10E						
	348° 166° 77 NM	FL 280 FL 245 Class A	10	↑	↓	
▲HEIDELBERG (HGV) VOR 264148.50S 0281659.39E						
	357° 176° 66 NM	FL 280 FL 245 Class A	10	↑	↓	
▲UTRUK 254032.98S 0274957.29E						

Route designator *Name of significant points* *Coordinates*	*Route* *MAG* *Track* *MAG* *DIST*	*Upper limit* *Lower limit* *Classification*	*Lateral limits* *NM*	*Direction of cruising levels*		*Remarks* *Controlling unit* *Frequency* *SATVOICE* *Logon address*
				Odd	*Even*	
1	*2*	*3*	*4*	*5*		*6*
	341° 159° 84 NM	FL 280 FL 245 Class A	10	↑	↓	
▲ORKAN 243230.00S 0265500.00E						
	338° 158° 16 NM	FL 280 FL 245 Class A	10	↑	↓	
▲RUDAS 241924.00S 0264406.00E						FAJA/FBGR FIR BDRY

UR987						
▲ERDAS 332450.10S 0191112.85E						
	022° 199° 122 NM	FL 280 FL 245 Class A	10	↓	↑	
▲NIEUWOUDTVILLE (NVV) VOR 312248.00S 0190416.00E						
	009° 187° 117 NM	FL 280 FL 245 Class A	10	↓	↑	
▲AGGENEYS (AGV) VOR/DME 292851.67S 0183344.97E						
	013° 193° 36 NM	FL 280 FL 245 Class A	10	↓	↑	
▲IBRUN 285324.75S 0182900.19E						
	013° 191° 84 NM	FL 280 FL 245 Class A	10	↓	↑	
▲KEBAT 273000.00S 0181800.00E						FAJA/FYWF FIR BDRY

UW61	

Route designator *Name of significant points* *Coordinates*	*Route* *MAG* *Track* *MAG* *DIST*	*Upper limit* *Lower limit* *Classification*	*Lateral limits* *NM*	*Direction of cruising levels*		*Remarks* *Controlling unit* *Frequency* *SATVOICE* *Logon address*
				Odd	*Even*	
1	*2*	*3*	*4*	*5*		*6*
▲PIETERMARITZBURG (PMV) VOR/DME 293852.13S 0302401.28E						
	295° 116° 87 NM	FL 460 FL 200 Class A	10	↑	↓	
▲NIDEM 293409.50S 0284409.10E						
	295° 115° 63 NM	FL 460 FL 200 Class A	10	↑	↓	
▲MASERU (MZV) VOR 293001.00S 0273226.00E						
	311° 131° 69 NM	FL 460 FL 200 Class A	10	↑	↓	
▲BRAM FISCHER INTL AIRPORT (BLV) VOR/DME 290602.38S 0261802.30E						
UW66						
▲GEORGE AIRPORT (GRV) VOR/DME 340026.46S 0222233.46E						
	055° 232° 187 NM	FL 280 FL 245 Class A	10	↓	↑	
▲OKSOP 311556.12S 0240832.74E						FAJA/FACA FIR BDRY
	052° 231° 61 NM	FL 280 FL 245 Class A	10	↓	↑	
▲PETRUSVILLE (PVV) VOR 302158.41S 0244142.06E						

UW81		
▲CHIEF DAWID STUURMAN AIRPORT (PEV) VOR/DME 335933.01S 0253620.75E		

Route designator *Name of significant points* *Coordinates*	*Route* *MAG* *Track* *MAG* *DIST*	*Upper limit* *Lower limit* *Classification*	*Lateral limits* *NM*	*Direction of cruising levels*		*Remarks* *Controlling unit* *Frequency* *SATVOICE* *Logon address*
				Odd	*Even*	
1	*2*	*3*	*4*	*5*		*6*
	032° 212° 50 NM	FL 280 FL 245 Class A	10	↓	↑	
▲EGTIM 330937.71S 0254125.96E						
	032° 210° 93 NM	FL 280 FL 245 Class A	10	↓	↑	
▲HOFMEYER (HMV) DVOR 313621.18S 0255038.40E						
	034° 213° 41 NM	FL 280 FL 245 Class A	10	↓	↑	
▲UTSEM 305604.65S 0255807.95E						FAJA/FACA FIR BDRY
	033° 212° 66 NM	FL 280 FL 245 Class A	10	↓	↑	
▲GEDOL 295037.49S 0261004.18E						
	032° 211° 45 NM	FL 280 FL 245 Class A	10	↓	↑	
▲BRAM FISCHER INTL AIRPORT (BLV) VOR/DME 290602.38S 0261802.31E						

UW95						
▲VICTORIA WEST (VWV) VOR 312349.00S 0230632.00E						
	074° 254° 19 NM	FL 460 FL 245 Class A	10	↓	↑	
▲OKBIL 311244.26S 0232351.31E						FAJA/FACA FIR BDRY
	074° 254° 84 NM	FL 460 FL 245 Class A	10	↓	↑	
▲PETRUSVILLE (PVV) VOR 302158.41S 0244142.06E						

Route designator *Name of significant points* *Coordinates*	*Route* *MAG* *Track* *MAG* *DIST*	*Upper limit* *Lower limit* *Classification*	*Lateral limits* *NM*	*Direction of cruising levels*		*Remarks* *Controlling unit* *Frequency* *SATVOICE* *Logon address*
				Odd	*Even*	
1	*2*	*3*	*4*	*5*		*6*
	069° 249° 113 NM	FL 460 FL 245 Class A	10	↓	↑	
▲BRAM FISCHER INTL AIRPORT (BLV) VOR/DME 290602.38S 0261802.31E						
	037° 216° 45 NM	FL 460 FL 245 Class A	10	↓	↑	
▲AVUBU 282242.79S 0263218.00E						
	036° 216° 23 NM	FL 460 FL 245 Class A	10	↓	↑	
▲WELKOM (WMV) VOR/DME 280005.74S 0263940.13E						
	051° 229° 104 NM	FL 460 FL 245 Class A	10	↓	↑	
▲GRASMERE (GAV) VOR 263053.63S 0274035.17E						